Crest - Portals

Release 1.3.0

Wave Harmonic

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CHAPTER

ONE

INTRODUCTION

The Portals package brings the Portals/Volumes/Tunnels feature to Crest 5. Portals restrict rendering of the water surface and volume to a mesh using either the front or back face. Volumes restricts rendering of the water surface and volume using both the front and back faces. Lastly, Tunnels removes water from within a mesh, which is useful for tunnelling through water.

This feature can be useful for aquariums and portholes. The promotional video shows the volume feature and then the portal feature (inverted) respectively.

1.1 Features

- Restrict water rendering to the front or back faces of a mesh
- Restrict water rendering to both the front and back faces of a mesh
- Remove water from within a mesh

CHAPTER

TWO

RELEASE NOTES

1.3.0

Changes

- Add Maximum Resolution and Texel Size controls for portal edge quality

Fixes

- Fix large portals not working

1.2.9

Fixes

- Fix meniscus rendering outside of fly-through volume

Optimizations

- Only use custom color texture if required

1.2.8

Fixes

- Fix null exceptions when enabling portals *HDRP*
- Fix underwater rendering inside inverted volume

Documentation

- Add basic usage instructions

1.2.7

Fixes

- Fix broken rendering with fly-through portal when enabled but not active

Optimizations

- Eliminate a GPU clear pass *BIRP*

1.2.6

Fixes

- Fix volume/caustics rendering inside of inverted volumes
- Fix refraction rendering incorrectly when inside inverted volume

1.2.5

Fixes

- Fix waterline when using a portal with a non-uniform bounds
- Fix underwater color mismatch between surface and volume when using inverted volumes

1.2.4

Fixes

- Fix inverted volume quality issues at eye level
- Fix inverted volume rendering on Windows *URP*
- Fix noticeable difference in underwater effect when camera crosses into a volume
- Fix meniscus on inverted volume intersections

Optimizations

- Remove no longer needed clear target pass for *Fly-Through* volume
- Simplify logic for inverted volumes

1.2.3

Fixes

- Fix *Fly-Through* volume breaking rendering for Unity 6 *URP*
- Fix *Fly-Through* volume causing the rest of the scene in the scene view not to render for DX11 *URP*

1.2.2

Changes

- Add tooltips to enum options
- Update GPU profiler / frame debugger markers naming conventions

Fixes

- Fix exceptions when enabling portals *HDRP*

1.2.1

Bump version number for compatibility.

1.2.0

Changes

- Add Tunnel mode for tunnelling through water (eg under rivers) including sample scene
- Add negative volumes via Volume mode plus Inverted. This is similar to Portal plus Inverted except it will render when the viewer is outside the portal

Fixes

- Fix Portal and Volume using the stencil buffer when they should not
- Fix Portal and Volume mode not working on PS5 platform

Full version history has been omitted for brevity. It can be found at Release Notes.

CHAPTER

THREE

PORTALS & VOLUMES

The water volume can be rendered from a provided mesh which will effectively become a portal (2D) or volume (3D). Change the Mode property to one of your choosing and set Geometry to a Mesh Filter (it will use its transform). This feature also clips the water surface to match. A common use case would be portholes (use Portal and Invert) or an aquarium (Volume).

Usage

Everything pertaining to portals is located at *Water Renderer* → *Portals*.

1. Add a MeshFilter to the scene and assign a mesh
2. Assign MeshFilter to *Water Renderer* → *Portals* → *Geometry*
3. Enable *Water Renderer* → *Portals* → *Enabled*

From here, choose a mode. The different modes are:

- **Portal:** Renders the water volume and surface from the geometry's front-face and behind it infinitely.
- **Volume:** Renders the water volume and surface from the geometry's front-face to its back-face.
- **Volume (Fly-Through):** Renders the water volume and surface from the geometry's front-face to its back-face - even from within the mesh.
- **Tunnel:** Removes water volume and surface using the mesh's front and back faces.

Limitations

- Only one mesh can be used at a time.